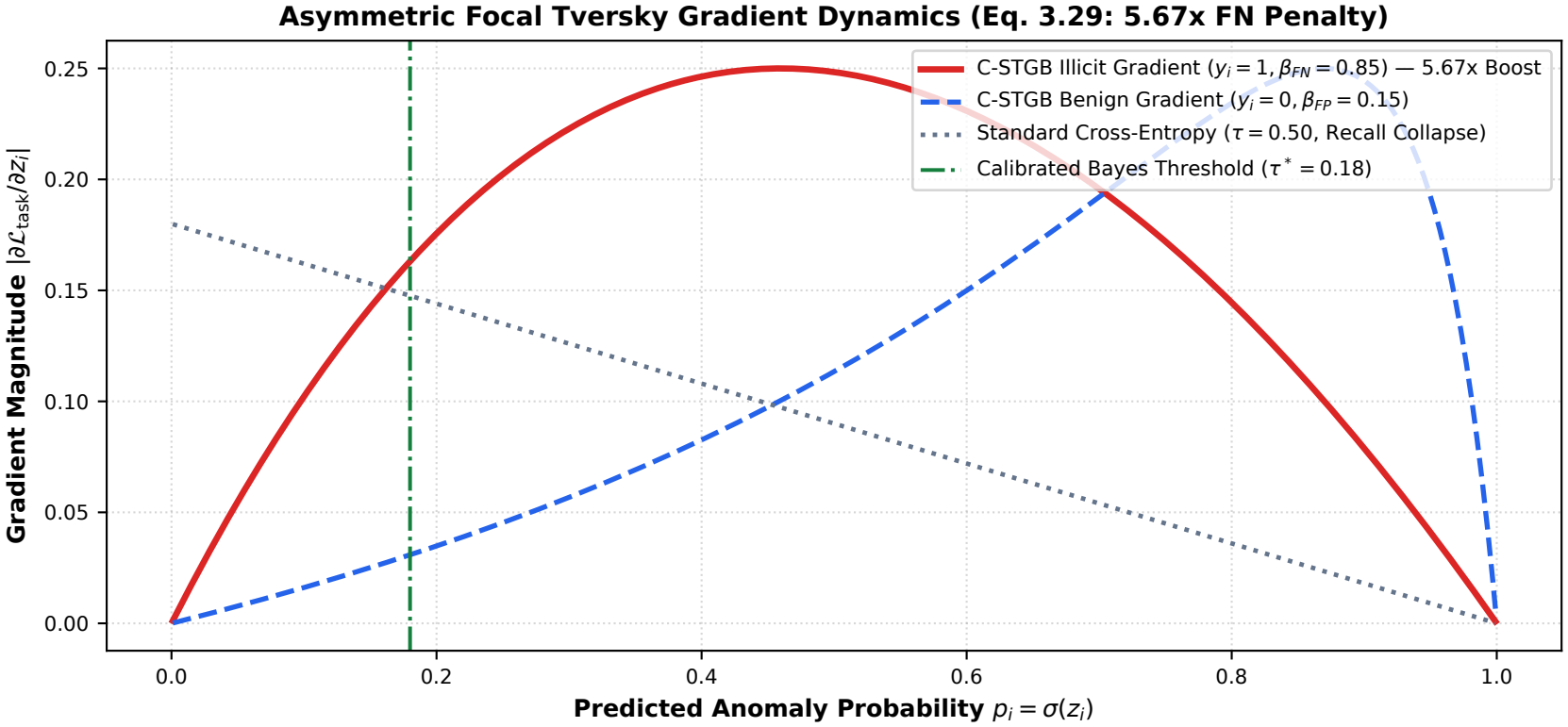
Asymmetric Focal Tversky Gradient Dynamics (Eq. 3.29: 5.67x FN Penalty)
C-STGB Illicit Gradient ($y_i = 1, \beta_{FN} = 0.85$) — 5.67x Boost
C-STGB Benign Gradient ($y_i = 0, \beta_{FP} = 0.15$)
Standard Cross-Entropy ($\tau = 0.50$, Recall Collapse)
Calibrated Bayes Threshold ($\tau^* = 0.18$)
Gradient Magnitude $|\partial \mathcal{L}_{task}/\partial z_i|$
0.25
0.20
0.15
0.10
0.05
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Predicted Anomaly Probability $p_i = \sigma(z_i)$